

Benchmark results compiled with -O2: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listr	handrolled.listr
countdown.shallow.10000	160.721	155.08	1248.605	4.566	144.537	287.854	269.619	2.32	88.416	197.481	1051.041	4.622	4.613	–	–	–
countdown.deep.10000	1273.499	567.558	2279.474	2702.808	146.691	276.331	163.353	2.423	166.586	194.517	3459.206	4.706	4.715	66.328	–	–
catch.shallow.10000	270.336	35.453	1110.792	20.746	1057.71	369711.9	246.784	16.209	73.185	834.045	831.301	20.71	20.469	–	–	–
catch.deep.10000	3178.598	35.704	6550.928	1678.98	1060.277	386628.4	173.678	16.241	235.604	832.657	840.909	20.218	20.642	167.878	–	–
local.shallow.10000	911.554	450.258	2162.567	4.891	540.036	370522.4	203.414	2.489	134.863	781.031	781.232	4.926	4.887	–	–	–
local.deep.10000	3764.781	447.846	5091.303	4.916	541.486	384970.3	205.02	2.475	213.864	788.28	787.625	4.902	4.855	204.311	–	–
nondet.shallow.32	1950.583	3222.383	7674.222	252.259	–	24440.537	757.724	80.046	906.215	1746.116	1734.524	219.591	242.432	–	359.659	360.846
nondet.deep.32	5373.616	5546.594	7986.356	2478.102	–	50161.05	755.348	80.544	623.973	5601.306	5603.594	242.718	241.624	1004.517	64579.975	62145.162

Benchmark results compiled with -O2: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listr	handrolled.listr
countdown.shallow.10000	6928.1%	6684.9%	53822.8%	196.8%	6230.4%	12408.4%	11622.3%	100.0%	3811.3%	8512.7%	45306.5%	199.3%	198.9%	–	–	–
countdown.deep.10000	52569.2%	23428.4%	94095.1%	111570.0%	6055.3%	11406.8%	6743.1%	100.0%	6876.6%	8029.5%	142793.7%	194.2%	194.6%	2738.0%	–	–
catch.shallow.10000	1667.8%	218.7%	6853.0%	128.0%	6525.5%	2280934.7%	1522.3%	100.0%	451.5%	5145.6%	5128.7%	127.8%	126.3%	–	–	–
catch.deep.10000	19570.9%	219.8%	40334.7%	10337.6%	6528.2%	2380507.1%	1069.3%	100.0%	1450.6%	5126.7%	5177.6%	124.5%	127.1%	1033.6%	–	–
local.shallow.10000	36623.0%	18080.8%	86884.2%	196.3%	21696.7%	14886270.5%	8172.5%	100.0%	5418.3%	31379.0%	31387.1%	197.9%	196.4%	–	–	–
local.deep.10000	152111.0%	18094.6%	205707.4%	198.6%	21878.0%	15554217.3%	8283.5%	100.0%	8640.9%	31849.4%	31823.0%	198.1%	196.2%	8254.9%	–	–
nondet.shallow.32	2436.8%	4025.6%	9587.2%	315.1%	–	30533.0%	946.6%	100.0%	1132.1%	2181.4%	2166.9%	274.3%	302.9%	–	449.3%	450.8%
nondet.deep.32	6671.7%	6886.4%	9915.5%	3076.7%	–	62277.9%	937.8%	100.0%	774.7%	6954.4%	6957.2%	301.3%	300.0%	1247.2%	80179.9%	77156.9%

Benchmark results compiled with -O2: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listr	handrolled.listr
countdown.shallow.10000	1760151	1600143	13361527	24	1121559	2087371	2480253	24	1040119	3441167	14241107	24	24	–	–	–
countdown.deep.10000	17760453	10400196	37363350	27121230	1136510	2088025	2000191	24	3920270	3442686	41443896	24	24	1200099	–	–
catch.shallow.10000	4069591	65593	7764882	65585	7846338	413420214	1687790	65584	723951	6081379	6081379	65585	65585	–	–	–
catch.deep.10000	16870411	66059	55153621	16165081	7864082	414233169	1527782	65584	3367870	6083939	6083909	65585	65584	1527782	–	–
local.shallow.10000	7205043	4793285	12629901	24	4326246	40950152	1669446	24	1029423	5513563	5513564	24	24	–	–	–
local.deep.10000	16005995	4793725	30232184	24	4341325	41040387	1589471	24	2549550	5515814	5515811	64	64	1589470	–	–
nondet.shallow.32	13454098	15889900	43492491	3606902	–	111260653	1531315	108343	1532747	22646045	22646045	1987327	2003982	–	3664087	3664093
nondet.deep.32	67560272	45646041	43569878	26810644	–	383226281	1531395	108343	1535147	68636406	68636439	1969094	2010670	7408468	469818614	469816735

Benchmark results compiled with -O2: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listr	handrolled.listr
countdown.shallow.10000	7333962.5%	6667362.5%	55673029.2%	100.0%	4673162.5%	8697379.2%	10334387.5%	100.0%	4333829.2%	14338195.8%	59337945.8%	100.0%	100.0%	–	–	–
countdown.deep.10000	74001887.5%	43334150.0%	155680625.0%	113005125.0%	4735458.3%	8700104.2%	8334129.2%	100.0%	16334458.3%	14344525.0%	172682900.0%	100.0%	100.0%	5000412.5%	–	–
catch.shallow.10000	6205.2%	100.0%	11839.6%	100.0%	11963.8%	630367.5%	2573.5%	100.0%	1103.9%	9272.7%	9272.7%	100.0%	100.0%	–	–	–
catch.deep.10000	25723.4%	100.7%	84096.2%	24647.9%	11990.9%	631607.1%	2329.5%	100.0%	5135.2%	9276.6%	9276.5%	100.0%	100.0%	2329.5%	–	–
local.shallow.10000	30021012.5%	19972020.8%	52624587.5%	100.0%	18026025.0%	1706642300.0%	6956025.0%	100.0%	4289262.5%	22973179.2%	22973183.3%	100.0%	100.0%	–	–	–
local.deep.10000	66691645.8%	19973854.2%	125967433.3%	100.0%	18088854.2%	1710002445.8%	6622795.8%	100.0%	10623125.0%	22982558.3%	22982545.8%	266.7%	266.7%	6622791.7%	–	–
nondet.shallow.32	12418.1%	14666.3%	40143.3%	3329.2%	–	102693.0%	1413.4%	100.0%	1414.7%	20992.2%	20992.2%	1834.3%	1849.7%	–	3381.9%	3381.9%
nondet.deep.32	62357.8%	42131.0%	40214.8%	24746.1%	–	353715.8%	1413.5%	100.0%	1416.9%	63351.0%	63351.1%	1817.5%	1855.8%	6838.0%	433640.0%	433638.3%

Benchmark results compiled with -O0: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listr	handrolled.listr
countdown.shallow.10000	784.412	508.358	1718.28	1700.222	722.01	872.4	1588.452	172.057	1612.417	843.552	2195.03	648.632	573.341	–	–	–
countdown.deep.10000	2869.136	954.393	2798.223	10447.962	728.489	866.319	1641.28	170.805	3252.877	1411.196	5118.972	5090.706	5848.544	1704.475	–	–
catch.shallow.10000	1047.003	301.265	2713.672	3305.22	2123.452	371047.4	939.531	343.794	1056.579	1608.369	1607.301	249.693	220.682	–	–	–
catch.deep.10000	8077.406	307.122	6894.306	15823.612	2164.041	385745.9	1025.487	346.353	2900.216	1614.48	1606.693	3820.494	4020.387	1109.514	–	–
local.shallow.10000	4446.012	655.506	2605.252	1400.88	1046.615	371884.5	609.596	292.655	640.404	1366.122	1366.67	347.051	325.565	–	–	–
local.deep.10000	13336.769	656.598	5516.988	5906.791	1058.198	386324.2	622.725	294.587	1601.967	1373.13	1379.691	1809.802	1584.84	629.876	–	–
nondet.shallow.32	7395.95	5879.847	13119.175	2163.648	–	29958.919	4083.792	5063.128	4122.258	2983.28	2993.454	2435.861	1952.288	–	7158.847	6647.222
nondet.deep.32	15514.419	7993.034	22624.338	10542.994	–	54016.725	4193.198	5584.325	9643.719	7739.15	7639.641	6724.841	6888.725	4473.978	83703.475	79284.5

Benchmark results compiled with -O0: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listr	handrolled.listr
countdown.shallow.10000	455.9%	295.5%	998.7%	988.2%	419.6%	507.0%	923.2%	100.0%	937.1%	490.3%	1275.8%	377.0%	333.2%	–	–	–
countdown.deep.10000	1679.8%	558.8%	1638.3%	6116.9%	426.5%	507.2%	960.9%	100.0%	1904.4%	826.2%	2997.0%	2980.4%	3424.1%	997.9%	–	–
catch.shallow.10000	474.4%	136.5%	1229.7%	1497.7%	962.2%	168136.9%	425.7%	155.8%	478.8%	728.8%	728.3%	113.1%	100.0%	–	–	–
catch.deep.10000	2630.0%	100.0%	2244.8%	5152.2%	704.6%	125600.3%	333.9%	112.8%	944.3%	525.7%	523.1%	1244.0%	1309.1%	361.3%	–	–
local.shallow.10000	1519.2%	224.0%	890.2%	478.7%	357.6%	127072.7%	208.3%	100.0%	218.8%	466.8%	467.0%	118.6%	111.2%	–	–	–
local.deep.10000	4527.3%	222.9%	1872.8%	2005.1%	359.2%	131141.0%	211.4%	100.0%	543.8%	466.1%	468.3%	614.4%	538.0%	213.8%	–	–
nondet.shallow.32	378.8%	301.2%	672.0%	110.8%	–	1534.6%	209.2%	287.0%	211.2%	152.8%	153.3%	124.8%	100.0%	–	366.7%	340.5%
nondet.deep.32	370.0%	190.6%	539.5%	251.4%	–	1288.2%	100.0%	133.2%	230.0%	184.6%	182.2%	160.4%	164.3%	106.7%	1996.2%	1890.8%

Benchmark results compiled with -O0: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	10640931	6080469	17281861	11920933	5202070	6268899	10801420	720158	11601191	7041755	19361713	4080229	2880411	-	-	-
countdown.deep.10000	38643716	14881331	41284523	63365168	5218157	6269552	11213144	720159	15371581	9444809	48965507	30882789	30882835	11365110	-	-
catch.shallow.10000	7913900	2571781	22351499	13075275	12013486	415979966	5433987	1462341	6393802	8161797	8161800	1673343	1593382	-	-	-
catch.deep.10000	33116297	2572597	57554507	64806100	12032265	416787691	5845706	1462343	10164146	8165099	8164589	16926380	18046462	5837709	-	-
local.shallow.10000	14981839	5771835	14008671	7688309	6726757	411514414	4008470	1767853	4648323	7193845	7193848	2327805	2327861	-	-	-
local.deep.10000	34184876	5772652	31611427	28490637	6742821	412321548	4020192	1767838	6818549	7197133	7196629	9128561	9048613	4012067	-	-
nondet.shallow.32	53827089	34532120	81643768	18238383	-	131919667	24814472	53824361	29682946	30881064	30881847	17396963	12390837	-	74123088	69116568
nondet.deep.32	137695451	64288394	200681315	63405443	-	403887170	26178261	53824623	42407641	80929169	80928666	43632108	39124520	28604455	543610378	539106931

Benchmark results compiled with -O0: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	1477.6%	844.3%	2399.7%	1655.3%	722.4%	870.5%	1499.9%	100.0%	1610.9%	977.8%	2688.5%	566.6%	400.0%	-	-	-
countdown.deep.10000	5366.0%	2066.4%	5732.7%	8798.8%	724.6%	870.6%	1557.0%	100.0%	2134.5%	1311.5%	6799.3%	4288.3%	4288.3%	1578.1%	-	-
catch.shallow.10000	541.2%	175.9%	1528.5%	894.1%	821.5%	28446.2%	371.6%	100.0%	437.2%	558.1%	558.1%	114.4%	109.0%	-	-	-
catch.deep.10000	2264.6%	175.9%	3935.8%	4431.7%	822.8%	28501.4%	399.7%	100.0%	695.1%	558.4%	558.3%	1157.5%	1234.1%	399.2%	-	-
local.shallow.10000	847.5%	326.5%	792.4%	434.9%	380.5%	23277.6%	226.7%	100.0%	262.9%	406.9%	406.9%	131.7%	131.7%	-	-	-
local.deep.10000	1933.7%	326.5%	1788.1%	1611.6%	381.4%	23323.5%	227.4%	100.0%	385.7%	407.1%	407.1%	516.4%	511.8%	226.9%	-	-
nondet.shallow.32	434.4%	278.7%	658.9%	147.2%	-	1064.7%	200.3%	434.4%	239.6%	249.2%	249.2%	140.4%	100.0%	-	598.2%	557.8%
nondet.deep.32	526.0%	245.6%	766.6%	242.2%	-	1542.8%	100.0%	205.6%	162.0%	309.1%	309.1%	166.7%	149.5%	109.3%	2076.6%	2059.4%